

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

Fig. 3: New channel on ThingSpeak platform

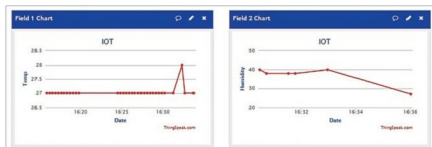


Fig. 4: Graphical view of humidity and temperature on ThingSpeak platform

Fig. 5: Channel Settings

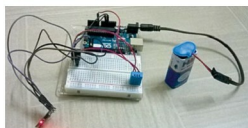


Fig. 6: Author's prototype

with your write API key:

```
String apiKey = "NTIHLKXET6VUWVF";
```

Next, substitute Host_Name and Password with your Wi-Fi name and Wi-Fi password in the two lines given below in the program (IoT.ino):

```
String Host_Name = "Jonah";
String Password = "2569696";
```

The program should be verified with your Wi-Fi setup. It uses DHT library. If DHT library is not present in your Arduino folder, download it from <https://github.com/adafruit/DHT-sensor-library>. To import DHT library in Arduino IDE, select Sketch→Import library→Add library→Select the library that you have downloaded.

Compile the sketch/program and upload it to Arduino MCU through Arduino IDE. Ensure that Wi-Fi modem and the Internet connection in your PC/smartphone are working properly.

Once sketch uploading is done, it will upload humidity and temperature values on ThingSpeak platform and you will be able to see it graphically in Private View window, as shown in Fig. 4. If you want to change channel or field name, you can change it from Channel Settings (Fig. 5). Author's prototype is shown in Fig. 6.

Other applications

Along with temperature and humidity sensor, other sensors like gas, voltage, current and energy can be used based on the requirement. Moreover, it is also possible to take further actions by controlling the actuators from the Internet once the sensor values are going above/below predetermined values. **EFY**

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